AI-DRIVEN DIGITAL TWIN FRAMEWORK FOR STRUCTURAL HEALTH MONITORING (SHM) OF A COMPOSITE AIRCRAFT WING SECTION

A MULTIDISCIPLINARY PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this multidisciplinary project report titled “AI-DRIVEN DIGITAL TWIN FRAMEWORK FOR STRUCTURAL HEALTH MONITORING (SHM) OF A COMPOSITE AIRCRAFT WING SECTION” is the bonafide work of “Aravindhan B., Anushka Mulik, Tanush Lad and Saood Bashir”, who carried out the project work under our supervisions. Certified further, that to the best of my knowledge the work reported herein does not form any other project report or dissertation based on which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

This paper demonstrates the design and experimental verification of a digital twin of a composite cantilever wing structural health monitoring. Test specimen was a 500 mm span tapered sandwich structure with steel face sheets and plywood core. 50% and 100% of span had 4 strain gages attached on the top and bottom surfaces to provide strain measurements in compression and tension; additionally, load cell was used for applied load and accelerometer for tip inclination measurement. Sensor data is captured through a ESP32 microcontroller and sent to a Python-based program to calculate bending stress, strain, tip deflection and tip angle via common beam theory and composite stiffness relations.

Load-deflection, load-strain, stress-strain, and load-angle characteristics are used to investigate experimental data. Data-driven methods were used to find inconsistencies in the normal strain signatures and estimate applied load from sensor outputs. Analytical solutions and actual stress distribution curves were compared to digital twin stress response curves. It is shown the system responds to structural behaviour in real-time and reproduces mechanical behaviour.

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Term / Symbol	Description
SHM	Structural Health Monitoring
AI	Artificial Intelligence
FEA	Finite Element Analysis
CFD	Computational Fluid Dynamics
ESP32	Microcontroller for data acquisition
ADC	Analog-to-Digital Converter
FFT	Fast Fourier Transform
RUL	Remaining Useful Life
HX711	Load cell amplifier module
I2C	Inter-Integrated Circuit communication
JSON	JavaScript Object Notation
FoS	Factor of Safety
Cp	Pressure Coefficient
F (N)	Applied force
m (kg)	Mass
g (m/s^2)	Acceleration due to gravity
δ (mm)	Tip deflection
θ (°)	Tilt angle
M (N·m)	Bending moment
σ (MPa)	Bending stress
ε ($\mu\varepsilon$)	Strain
E (Pa)	Young’s modulus
I (m^4)	Second moment of area
EI (N·m^2)	Flexural rigidity
L (m)	Wing span length
x (m)	Spanwise position
c (m)	Distance from neutral axis

b (m)	Width of wing section
t (m)	Thickness of section
a_x (m/s²)	Acceleration in x-direction
a_γ (m/s²)	Acceleration in y-direction
f (Hz)	Frequency
K (N/mm)	Structural stiffness
G1, G2	Top surface strain gauges
G3, G4	Bottom surface strain gauges
σ_{max} (MPa)	Maximum stress
δ_{theo} (mm)	Theoretical deflection
δ_{exp} (mm)	Experimental deflection

LIST OF EQUATIONS

CHAPTER 1

INTRODUCTION

1.1 Structural Behaviour of Aircraft Wings

Aircraft wings are designed as primary load-carrying components and are subjected to aerodynamic loads distributed along the span. As the wings generate aerodynamic loads, these loads in turn develop bending moments, shear forces, and thus corresponding stresses throughout the span. For the specific case of the cantilever wing structure discussed, the bending moment varies along the span and reaches maximum value at the root of the wing and diminishes to zero at the tip. The stress distribution along the span of a cantilever wing is therefore a function of spanwise load and bending moment distribution and is analyzed using the assumption of beam theory. In most of these analyses Euler-Bernoulli beam assumptions are applied for slender structures where the beam bending stress is given by:

$$\boldsymbol{\sigma} = \frac{\boldsymbol{Mc}}{\boldsymbol{I}} \qquad \textbf{Equation 1}$$

The tip deflection of the wing can also be calculated as a function of load, span of the wing, flexural rigidity of the beam etc. Jones [1], Megson [2], Bruhn [3] provide details for the theoretical calculation of behavior of composite as well as metallic wing structures including that for sandwich construction as shown in figure 1.1.

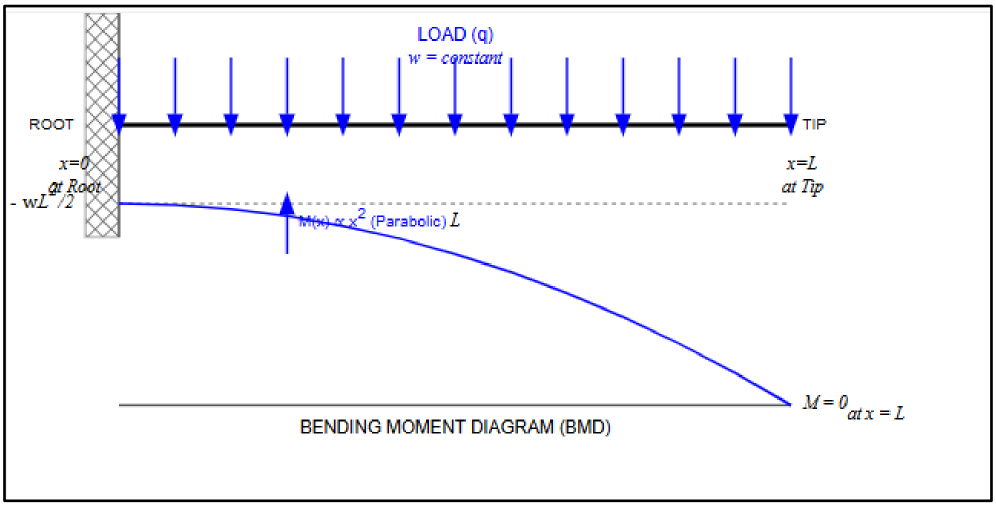


Figure 1.1: Cantilever Wing Showing Load Distribution and Bending Moment

In the present analysis the wing is designed as tapered composite sandwich structure composed of steel face sheets and plywood core. Such a configuration's rigidity will be a sum of its constituents, hence evaluated using parallel axis theorem to calculate equivalent flexural rigidity (EI). As per sandwich panel theory [5] the faces of the structure are responsible for carrying the bending stresses and core to resist shear forces between them. Thus this construction becomes stiff in bending.

1.2 Limitations of Simulation-Based Structural Analysis

CFD simulation tool is widely used for predicting the aerodynamic loading that aircraft wing has to carry and FEA is applied for determining its behavior against these loads. CFD calculates pressure distribution and the resultant lift force while FEA utilizes boundary conditions set in a numerical model to compute the resultant deformation and stresses. The limitation of these simulation techniques arise from ideal assumptions of boundary conditions as well as for the materials and the connections.

The results that are derived from these simulations must therefore always be compared and validated against experimental findings. These differ in practice due to several reasons like; Non-ideal boundary condition (the joint of the wing and body are not fixed completely). Also, due to manufacturing errors, or the imperfections that arise due to non-ideal construction. For instance clamp compliance leads to rotational flexibility thus reducing stiffness, etc.

1.3 Experimental Measurement of Structural Response

Experimental structural analysis techniques can measure actual deformation of a structure through sensors like strain gauges, load cells and accelerometers. Strain gauges are electrical devices that measure strain on a surface based on resistance change. The measured strain value is proportional to actual deformation. The relation between the strain and stress is given by:

$$\boldsymbol{\sigma} = \boldsymbol{E} \cdot \boldsymbol{\epsilon} \qquad \textbf{Equation 2}$$

Where, E is the modulus of elasticity of the material. Standard practices to perform strain measurement including calibration are described in ASTM E251 and in the relevant documents [6-8].

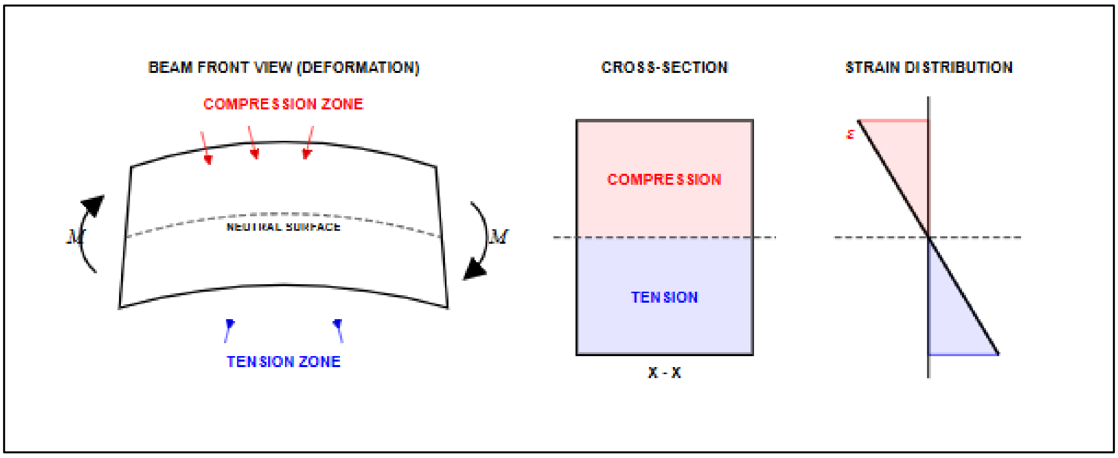


Figure 1.2: Strain Distribution across Wing Thickness

In a cantilever setup the strain value will be zero at the midspan or in the neutral axis and maximum at the extremities. It will be tensile in nature on the lower and compressive in nature on the upper surface and it varies linearly along the thickness. If strain gauges are placed at different locations and on both faces of the wing and measured, then strain at each point will be recorded along the span. A load cell provides a direct measure of the applied force on the structure. Accelerometer measures acceleration which through its components will indirectly be used to calculate angular displacement. The above Figure 1.2 shows the bending of beam which shows the compression and tension in the beam.

A limitation of experimental measurement is that signals from each sensor are output in terms of either voltage or ADC value. These are not direct measures of engineering parameters and therefore have to be converted to strain, stress, moment or deflection using a suitable analytical relationship.

1.4 Need for Integrated Structural Evaluation

In traditional techniques simulation and experimental analysis results are treated separately. FEA can predict stress and deformation but experiment generates only isolated data points. There is no direct framework for integrating between those results that can be used to compare structural behaviour.

The results from experimental measurement, as indicated before, require an extensive analysis before it can be converted into useful structural information and can only be viewed after completion of the loading process. Hence for applications like structural monitoring it becomes imperative to process experimental sensor outputs in real time into structural response and to compare with simulation based prediction.

A significant drawback in analysis of structural components is the lack of a direct method to translate measured strain and load to structural response that can further be correlated with experimental setup, analytical computation and finite element method simulations.

1.5 Digital Representation of Structural Response

The present system, therefore, involves measuring strain signals from the wing structure. The wing structure will be instrumented with 4 strain gauges at 50% and 100% span along both the upper and lower surfaces, along with a load cell to measure the applied force and an accelerometer to record instantaneous dynamics. These values from sensors are fed into an ESP32 microcontroller for processing. The ESP32 samples the sensor readings and sends them over serial port to be processed. Using beam theory relations, these measured signals are then converted into dynamic behaviour of structure which include bending moment, stress, strain value on outer and inner fibres of wing, the tip deflection, and rotation angle of the structure. The Figure 1.3 shows the sensors flow for data acquisition.

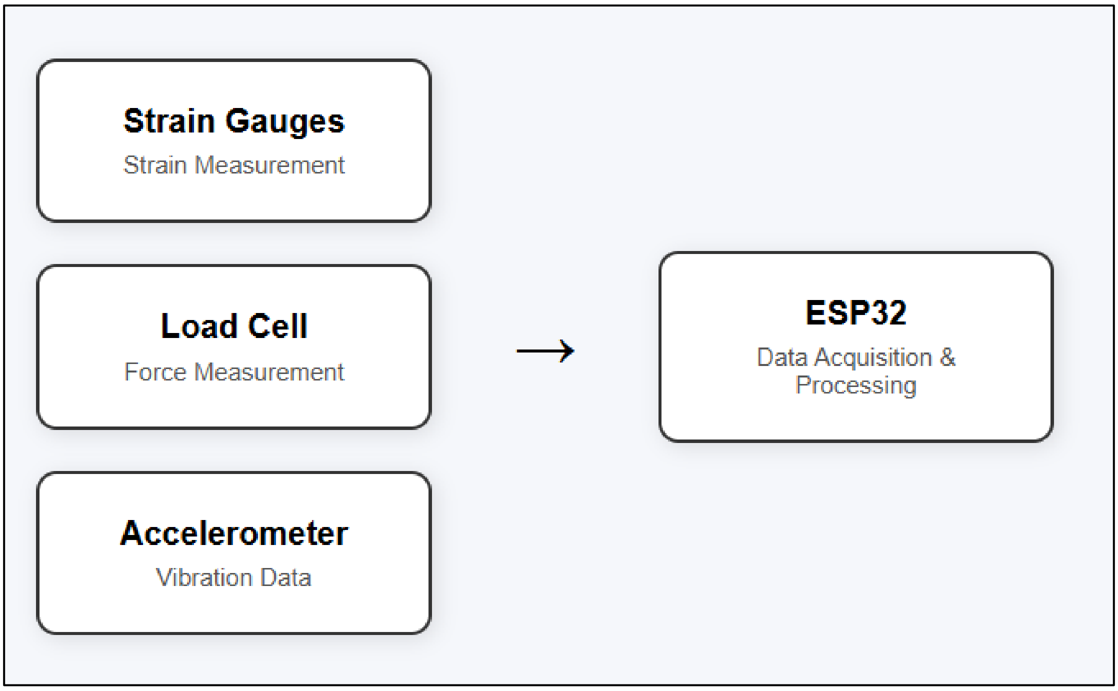


Figure 1.3: Conceptual Representation of Sensor-Based Structural Monitoring System

All these data would be continually stored and displayed. An attempt to visualize behavior of the wing by representing these numerical values as numbers on the screen, plots of load vs deflection and stress vs strain, and also 3D plot for the wing deformation shape, are made in this system.

1.6 Data-Driven Evaluation of Structural Behaviour

Besides the analytical approach of converting measurement results into structural behaviours and vice versa, This system also incorporates a data driven approach of predicting them using data derived from experiment. Algorithms such as Random Forest regression [16] for predicting load using measured strains at specific locations and Isolation Forest [17] are being implemented for anomaly detection, that may be related to abnormal load or stress condition or any fault in the structure or instrumentation.

These models will take readings from different strain gauge outputs and analyze them in different combinations to make a prediction. A final health score will then be computed based on these readings and predictions and used to classify structural condition (e.g. Good health, overload, under-load or faulty) into three possible ranges.

1.7 Scope of the Present Work

This work will entail the designing and fabrication of a composite wing structure which is then instrumented with the sensors and connected to an electronic system that acquires signals and processes them into structural behaviour metrics. Analytical approach to predict structural behaviour from measured strain and the results are then compared with the FEA prediction.

CHAPTER 2
LITERATURE SURVEY

2.1 Mechanics of Composite Sandwich Wing Sections

Jones et al. [1], *Kassapoglou et al.* [4], *and Allen et al.* [5] noted that, the popularity of composite sandwich sections for aeronautical applications stems mainly from their outstanding stiffness-to-weight ratio which far exceeds that of homogeneous constructions [1,4,5]. In typical composite sandwiches, two relatively thin, high-modulus face sheets are separated by an elastic low-strength core material and structural response differs greatly from a solid beam. It is widely reported that the majority of bending stresses are carried by the face sheets; the core mainly serves to keep the faces apart, thereby expanding the second moment of area, and increases bending stiffness. Therefore, inexpensive, low-strength core materials may be used with only a slight sacrifice in bending stiffness. It is typical for classical analytical approaches to reduce a composite section to an equivalent homogeneous section through an equivalent material modulus substitution method [1,4]. However, this method presupposes perfect bond between layers and homogeneous properties. Several papers point to a high sensitivity of the structural behaviour to the manufacturing process. Small deviations in bonding quality or laminate layup could cause big discrepancies between calculated and experimental stress distribution and stiffness [1,5]. For an aircraft wing, sandwiches are particularly attractive in increasing the bending stiffness without adding a weight penalty. However, their structural behaviour when used in tapered sections, subject to more practical loading and boundary conditions, is inherently more difficult to analyse [2,3].

2.2 Classical Beam Theory and Tapered Wing Analysis

Timoshenko et al. [9], *Gere et al.* [10], *Megson et al.* [2], and *Bruhn et al.* [3] explained that Classical beam theory is the basis for most analysis methods for cantilever wing structures [9,10]. Euler-Bernoulli beam theory relies on the assumption that plane sections are to remain plane and normal to the neutral axis during deformation; this is particularly useful for easy analytical assessment. For a cantilever beam subject to a point load at its free end, deflection is proportional to:

$$\delta = \frac{FL^3}{3EI} \qquad \textbf{Equation 3}$$

This is often used as a benchmark and compared to experimental and numerical results [9,10]. However, in many analyses of tapered wing structures this theory is often considered to be inadequate as there is constant variation of chord length and section properties along the length of the wing [2,3] and cannot be treated as having constant section properties. As such the wing section is analyzed as being made up of several sections of differing geometries and rigidity. Finite Element Analysis (FEA) is often used to perform such analysis, and can easily model varying geometry and material distributions [2,3]. From research, it can be noted that both numerical and analytical solutions are very sensitive to the boundary conditions. It is a common finding that in lower load cases FEA under-predicts beam deflection because of the perfect fixing assumption and that in a practical situation the root has compliance [3,10]. As the load on the beam increases, so the influence of root compliance reduces relatively.

2.3 Experimental Stress Analysis and Sensor Calibration

Dally et al. [6], ASTM E251 [7], and ASTM E837 [8] described the use of many experimental techniques for the purpose of characterizing the true performance of structures. Strain gages are probably the most commonly used measurement device in this field, and a significant amount of literature has been dedicated to how they should be installed and calibrated [6-8]. If the surface to which the gages are bonded has not been properly prepared or the adhesive is not properly applied, large measurement errors can be introduced. Furthermore, if the gage is not correctly oriented, measurement errors will occur. The electrical output of the strain gage has to be converted to strain through a calibration factor that is typically obtained empirically by applying a known load that has been previously characterized by a calibrated load cell [6,7]. The measurement error will be a direct result of the calibration error in the load cell or variability in load application. Accelerometers can also be used to monitor vibration, as well as obtain estimates of angular displacement under static conditions [25]. Filtered outputs of accelerometers are often used to characterize structural behavior due to sensor noise and drift.

2.4 Digital Twin Paradigm for Aerospace Assets

Glaessgen et al. [14], *Grieves et al.* [15], *Boschert et al.* [13], *Tao et al.* [11] and *Fuller et al.* [12] described the digital twin; which is typically a up to date computation model of a physical component or system [11-15]. In the area of aerospace the role of digital twin is used to predict structures behavior when under operational loads. The most commonly described digital twin systems involve three main elements: the physical system, the sensor system and the model system [11-15]. When it is concerning a wing component, the physical system refers to the wing itself and the sensors will be used to monitor the structural behavior, in order to update the model system. As it can be noted in the literature it can be inferred that most systems use a large number of previously computed simulations [12-15]. However some authors state that due to environment, manufacturing uncertainties and operating conditions, the reliability of models is decreasing [11-12] and therefore a number of recent publications suggest coupling simulation results and real-time sensor data to gain prediction reliability [11-15]. Sensor reliability and calibration are the main challenge in this case, since data from sensors will directly influence the quality of the digital system.

2.5 Real-Time Data Acquisition and IoT Frameworks

Fuller et al. [12], *Ogunleye et al.* [36] and recent SHM sensor studies [38] dscussed aforementioned, the recent trend of embedded systems has enabled real-time acquisition and communication of structural sensor data [12,36,38]. MCUs like ESP32 are quite commonly utilized since they support wireless communications and they can connect to numerous sensors concurrently. In most designs, the acquired sensor signals are read from sensors with a predefined sampling period and the data are transferred to remote system by communications such as HTTP or MQTT depending on requirements regarding delay, reliability and implementation ease. Synchronization is universally addressed as a requirement in real-time monitoring systems as signals from numerous sensors have to be acquired and process at the same time in order to provide accurate reflection on structural response [12,36]. Communication reliability is another significant issue in wireless communication; it is very common that the transmission packet is dropped or delayed in wireless networks and thus measurement is not reliable. Real-time structural monitoring systems therefore must take into account the effects of communications errors. In most papers, computational processing on the data is executed by the computer and sensor signals are converted into engineering values based on analytical equations and then plotted for monitoring and analyses [12,36].

2.6 Structural Health Monitoring (SHM) and Feature Extraction

Farrar et al. [18], Sohn et al. [19], Worden et al. [20], and Boller et al. [21] points out that SHM consists of processing data from sensors in order to determine the state of a structure [18–21]. Older methods based on threshold detection compare the measurements with expected values. Newer SHM strategies instead analyze feature vectors that can contain parameters such as strains and their changes, vibration modes, trends of the loads and responses, etc. [18,19]. Feature vectors contain more information than raw measurements and can effectively capture structural alterations. Methods based on time-series are commonly employed to highlight the deviations of the system's response from its normal behavior [18–21]. When using a network of sensors the data from different sources are integrated in order to assess the condition of the structure. Multi-sensor SHM systems are thus of particular relevance for aerospace applications due to the high complexity of the structures [21,29] and the lack of information obtainable by a single sensor.

2.7 Machine Learning for Load Prediction and Regression

Breiman et al. [16], *Scarselli et al.* [31], and *Azad et al.* [33] observed that machine learning algorithms are widely applied to Structural Monitoring purposes to approximate values that are not easily to be obtained directly [16,18,31,33]. Regression algorithms, specifically, have become widely used to estimate structural parameters from the measurements obtained through sensors. One of the parameters to be estimated could be the load that acts on the structure based on the sensor measurements (strain values). Random Forest Regression algorithm by Breiman [16] is one of such widely used regression algorithms that performs well even in the presence of nonlinear relationship between inputs and outputs andnoisy datasets. An important advantage of machine learning algorithms is that they do not need any explicit analytical equations to relate sensor values to unknown values. In fact, these relationships are learned directly from experimental data. However, previous studies showed that performance of any machine learning algorithms are very sensitive to the size and quality of training data [16,31,33], thus many machine learning techniques need to be employed along with physics-based and analytical modeling.

2.8 Anomaly Detection and Predictive Health Metrics

Liu et al. [17], *Worden et al.* [20], *Doebling et al.* [26], *Rytter et al.* [27], *Fan et al.* [28], and *Farrar et al.* [29] observed that anomaly detection algorithms have been used to detect structural anomalies which can signify any type of structural degradation (i.e. Damage, overloading, and sensor failure). Outliers can be detected using an Isolation Forest algorithm which is very suitable for detecting anomalies within sensor data sets [17]. Any changes in the baseline structural response are therefore viewed as anomalies and potential evidence of changes to the structural behavior. Health indicators are computed by combining readings from several different sensors and these readings then undergo analysis using a combination of different analytical and machine learning approaches [18-21]. The derived health indicator reflects the overall health status of the structure. Many of the potential problems within anomaly detection are the distinction between true structural anomalies and measurement noise [20,26-30]. Sensitive systems may generate an abundance of false positives, while a highly tolerant system may be incapable of detecting a true anomaly. Detection algorithms therefore must be fine-tuned appropriately for reliable structural health monitoring.

2.9 Literature Summary

According to the literature that reviewed, Beam theory, FEM and experimental stress analysis are commonly adopted in the investigation on the behavior of aircraft wing, namely the behavior of Deformation, Stress and Rigidity. As for the analytical approach on the wing with cantilever boundary, Megson et al. [2], Bruhn et al. [3] and Timoshenko et al. [9] are cited, and Jones et al. [1], Kassapoglou et al. [4] and Allen et al. [5] introduced analysis on stiffness of composite sandwich structure, etc. Strain gauge, load cell and accelerometer were common measuring instruments in experimental stress analysis, and were mentioned in Dally et al. [6] and ASTM standards [7,8]. Calibration and accuracy in the use of these measuring instruments also discussed in these papers.

Recently, much concern is paid to digital twin and SHM systems on aircraft. Tao et al. [11], Fuller et al. [12], Glaessgen et al. [14] and Grieves et al. [15] mainly describe the concept of digital twin and integration between the physical and digital system. Correspondingly, ML methods such as Random Forest regression, Isolation Forest, are used for load estimation and anomaly detection in SHM systems. Generally, more focus are laid on the integrated system rather than single analysis or experimental approaches.

2.10 Literature Gap

Although a lot of analytical, FEM and SHM analyses as well as the conception of digital twin [2,3,9,10] are studied on the airplane wing, some parts are not yet implemented experimentally. The majority of the analytical and FE researches on an airplane wing are focused on ideal structure with a simple boundary condition, while the experimental researches are carried out with separated measurement systems rather than the actual processing of structural response. Besides, the concept of digital twin related research relies mainly on a simulation-based study which ignores the dynamic response of structure using experimental feedback [11-15].

Furthermore, the measurements are almost always about individual strain values rather than application in an engineering field such as real-time reading of deflection, stress, structural health status, etc. The vibration based damage detection is the major area covered in SHM research, it is performed with the employment of a highly expensive aerospace sensor system and lacks of the concept of an integrated low-cost embedded sensing system that connects experiment, analytical validation, simulation and machine learning together [18-21, 29-38].

The use of distributed multiple sensors on a tapered composite sandwich structure airplane wing which transmits data to a real-time processing system via an embedded controller such as ESP32 is still new and not many researches are performed on that. This thesis aims at integrating the beam theory, experimental investigation, FEA and machine learning together for a structural evaluation frame.

2.11 Objectives

The principal aim of the current work is to design, manufacture and test a tapered aircraft wing section, that will exhibit significant elastic deformation under a static cantilever loading condition. The aim is to analytically calculate, through finite element analysis and experimentally determine structural performance characteristics of the wing section.

A further aim is to construct a sandwich construction wing configuration with stainless steel 304 as face sheets and plywood as a core material to achieve a satisfactory blend of stiffness and detectable elastic deformation and investigate the behaviour of strain, stress, bending moment and deflection along the span of the wing by applying an array of strain gauges, load cell and an accelerometer.

An additional aim of this work is to analytically determine the bending moment, deflection and strain using beam theory and through Finite Element Analysis and compared with experimental data. CFD is carried out to obtain aerodynamic load values expected at different configurations of flight.

An additional aim is to use machine learning algorithms to predict the loads acting on the wing section and detect any anomaly within the wing from the measured strain value. Ultimately this research aims to formulate a united analytical, numerical, experimental and data driven framework for static cantilever wing analysis.

CHAPTER 3
DESIGN AND ANALYSIS OF AIRCRAFT WING

3.1 Designing and manufacturing of the wing

In the present work, the main aim is to develop a structurally representative aircraft wing structure that can generate measureable elastic deformation when static loads are applied. The wing must be sufficiently stiff to be repeatable during experimentation. For purposes of comparison with analytical models, Finite Element analysis and experimental stress analysis, the geometry had to support all of these and the other criteria were manufacturability, ability to mount sensors, and ability to withstand cantilever loading without twisting.

During operation aircraft wings mainly act as cantilever beams, being subject to bending moments varying from a maximum at the wing root to zero at the tip. As a result the one face of the wing develops tensile strain and the other compressive strain. In this present work as the experimental investigation is focused on to determine the bending behavior it was necessary for the structure to develop measurable strains and be completely elastic within the test range.

The wing geometry chosen for use in the present work is a downscaled scaled up version of a tapered cantilever wing discussed in textbooks such as Aircraft Structures by Megson et al. [2] and Airplane Structures by Bruhn et al. [3]. Generally for practical aircraft wings tapered plan forms are used as this reduces the weight and structure has higher stiffness at the root where bending is greatest due to decreasing chord length along the span. However, in the present work, the scale was downscaled and was to be a structure used on a laboratory scale, and loaded statically.

The final wing structure chosen was having a span of 500mm, a root chord of 200mm, tip chord of 125mm and an overall thickness of 30mm. This size was chosen after several iterations of design such that it generated a measureable amount of deformation without the wing becoming too flexible. The reducing chord along the span resulted in a variation of sectional stiffness along the length of the wing allowing determination of variation of bending response from root to tip. All the dimensions for the final wing can be seen in Table 3.1. Figure 3.1(a) and (b) show the tapered wing geometry and relative proportions respectively.

Table 3.1: Wing Dimensions specifications

Parameter	Value
Half Span	500 mm
Root Chord	200 mm
Tip Chord	125 mm
Thickness	30 mm
Wing Configuration	Tapered Cantilever
Structural Type	Sandwich Structure

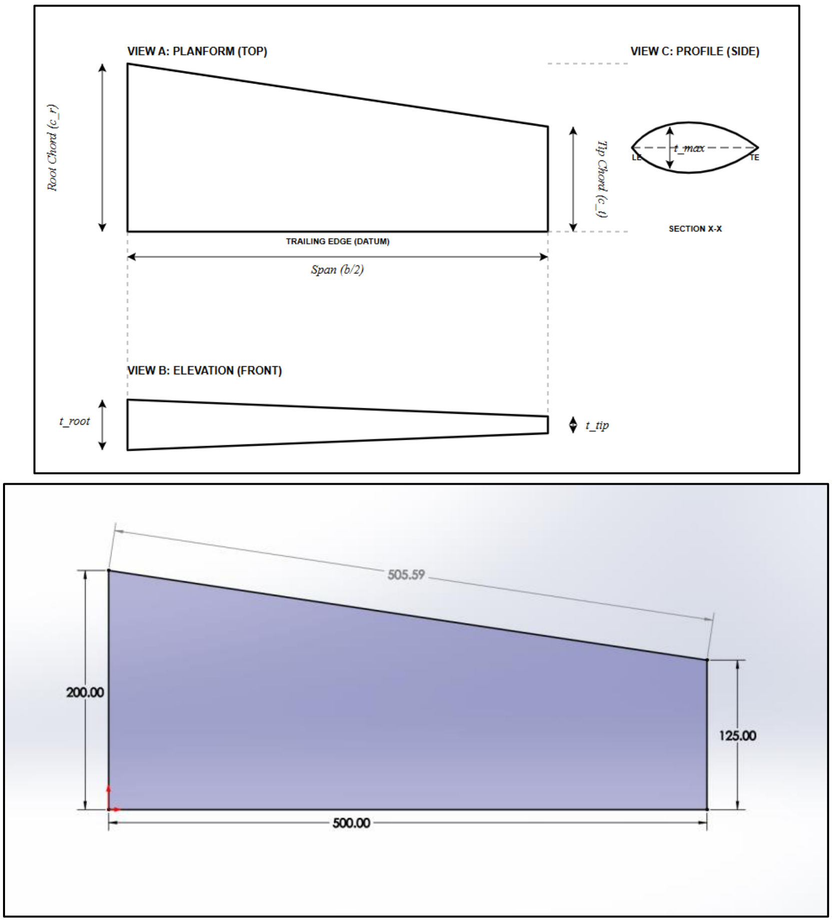


Figure 3.1: Tapered wing Geometry showing span, root chord, tip chord and thickness

The structural design evolved through multiple prototype configurations before the final arrangement was selected. The first prototype consisted entirely of stainless steel 304 sheet material and was initially developed for modal and vibration-based investigation. Since stainless steel possesses relatively high stiffness compared to the geometry of the structure, the wing exhibited extremely small measurable deformation under practical loading conditions. From the initial inspection, the structure acted almost like a rigid body within the excitations range we could test within. It was not found possible to test the structure in a range where the resulting displacements would be suitable for modal testing. The current paper is concerned with analysis of measurable static deformation and distributed strain evaluation, so no further work on the fully metallic design was done. The first SS304 wing structure prototype constructed at this stage is shown in Figure 3.2.

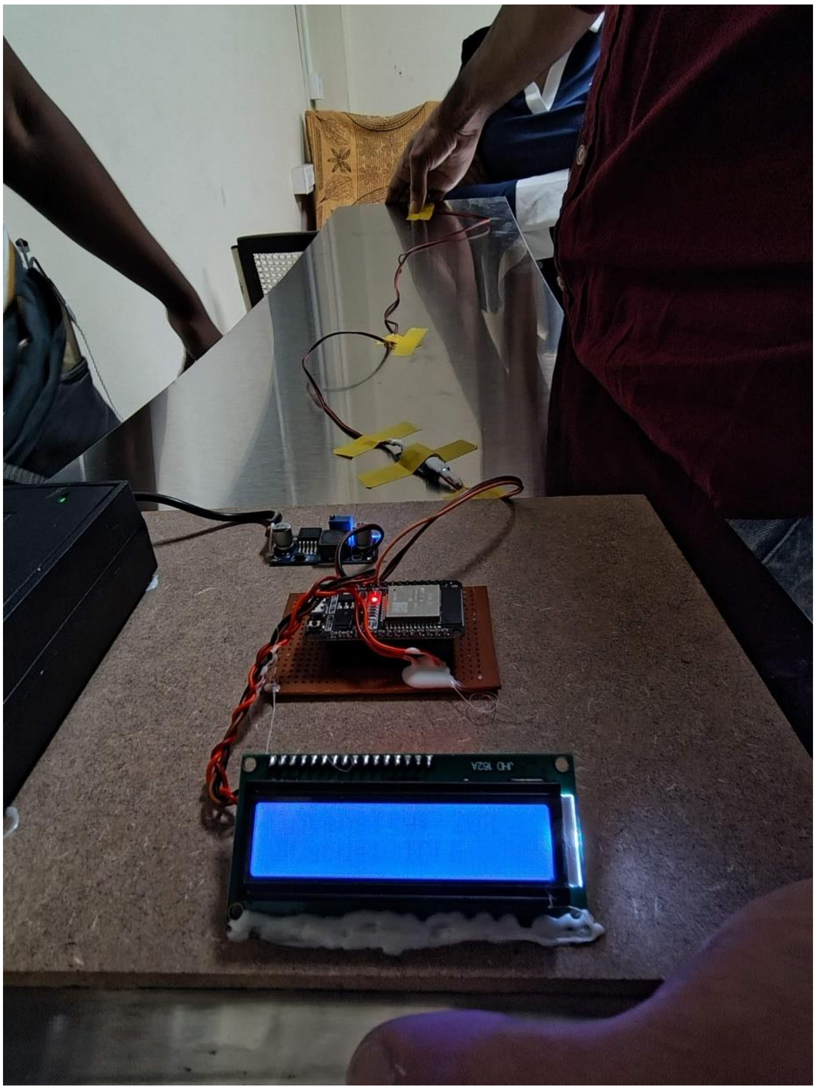


Figure 3.2: Initial SS304 Wing Prototype Developed for Modal Investigation

The second prototype configuration was therefore developed. To improve flexibility and to produce large measurable deflection, it comprised of acrylic ribs and spars covered by a 5mm foam sheet skin. The half span of this wing was approximately 1 m and again intended to develop a large static deflection. The acrylic structure provided flexibility and the foam sheet provided surface contour and continuity. Tests were carried out and this structure exhibited much larger deflection but closer inspection shows shortcomings, on examination it was apparent that there were several problems with this structure, though the deflection was readily measurable it was heavily influenced by local flexibility leading to insufficient torsional stiffness of the wing and visible deflection of the foam skin, inconsistent stiffness along the length in the ribs and spars. Multiple loading cycles also resulted in inconsistent deflection, rendering the structure unsuitable for repeatable strain analysis or comparison with analytical work. The findings from the initial wing designs have been summarized in Table 3.2, the structure of the acrylic rib-spar wing is also shown in Figure 3.3.

Table 3.2: Observations from initial prototype configurations

PROTOTYPE	CONSTRUCTION	STRUCTURAL BEHAVIOUR	LIMITATION
PROTOTYPE 1	SS304 Sheet Structure	Very high rigidity	Negligible measurable deformation
PROTOTYPE 2	Acrylic Rib-Spar with PVC Foam Skin	Large deformation	Excessive flexibility and local compliance

Figure 3.3: Acrylic Rib-Spar Wing with Foam Skin Used for Preliminary Structural Investigation

Using what was gained from the prior designs, the final configuration of the wing was developed into a short, tapered sandwich beam structure. The purpose of the final configuration was to maintain a balance between measureable elastic deformation and adequate bending stiffness. The final design consists of stainless steel 304 sheets used as the top and bottom face sheets and plywood used as the sandwich core.

With the use of a sandwich beam design, significant structural benefits over the previous configurations are apparent. The overall second moment of area of the beam is increased, because the steel face sheets are placed a distance from each other. Since the bending stiffness of a beam is the product of Young's modulus and second moment of area, this greatly increases the flexural stiffness. The steel face sheets carry predominantly the compressive and tensile stresses during bending and the plywood simply acts as an in-between layer, and increases the stability.

The fabrication was done by preparing the plywood core in a tapered shape, then cutting stainless steel sheets into the required planform and bonding them to the top and bottom faces of the plywood beam. Both face sheets were carefully aligned throughout the length of the beam to distribute stiffness uniformly and to avoid local stress concentrations. Upon completing the assembly, the wing root section was reinforced in anticipation of the cyclic loading expected during experimental analysis. Figure 3.4 shows the completed sandwich wing structure.

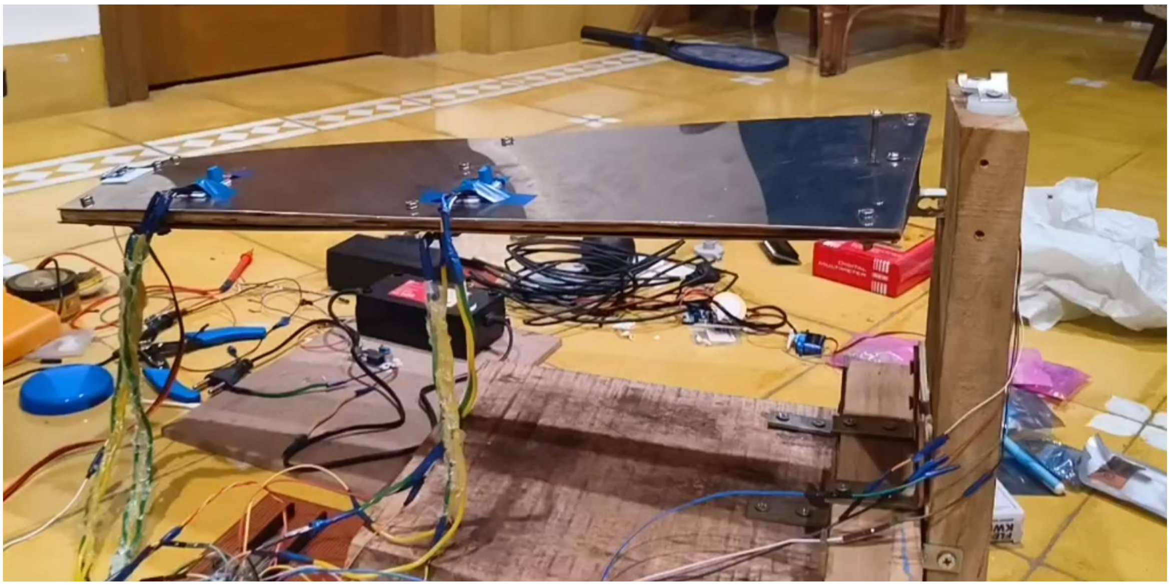


Figure 3.4: Fabricated Sandwich Wing Structure with SS304 Face Sheets and Plywood Core]

The structural behavior of the wing is measured through strain gauges that are mounted at two distinct spanwise locations, on the top and bottom faces of the wing. This type of arrangement allows the viewing of compressive strain at the top and tensile strain at the bottom during cantilever loading of the wing simultaneously. A load cell is used to monitor the magnitude of the load applied, and an accelerometer is used at the tip of the wing to determine the angular displacement during bending of the wing. The arrangement of the sensors and their function are outlined in Table 3.3, and the physical location of strain gauges are identified in Figure 3.5: Strain Gauge Placement on Upper and Lower Wing Surfaces.

Table 3.3: Sensor Placement and Structural Purpose

SENSOR	LOCATION	STRUCTURAL PURPOSE
STRAIN GAUGE 1	Upper Surface 50% from Root	Compression strain measurement
STRAIN GAUGE 2	Lower Surface 50% from Root	Tensile strain measurement
STRAIN GAUGE 3	Upper Surface 100% from Root	Compression strain measurement
STRAIN GAUGE 4	Lower Surface 100% from Root	Tensile strain measurement
LOAD CELL	Wing Root	Applied load measurement
ACCELEROMETER	Wing Tip Region	Angular deformation estimation

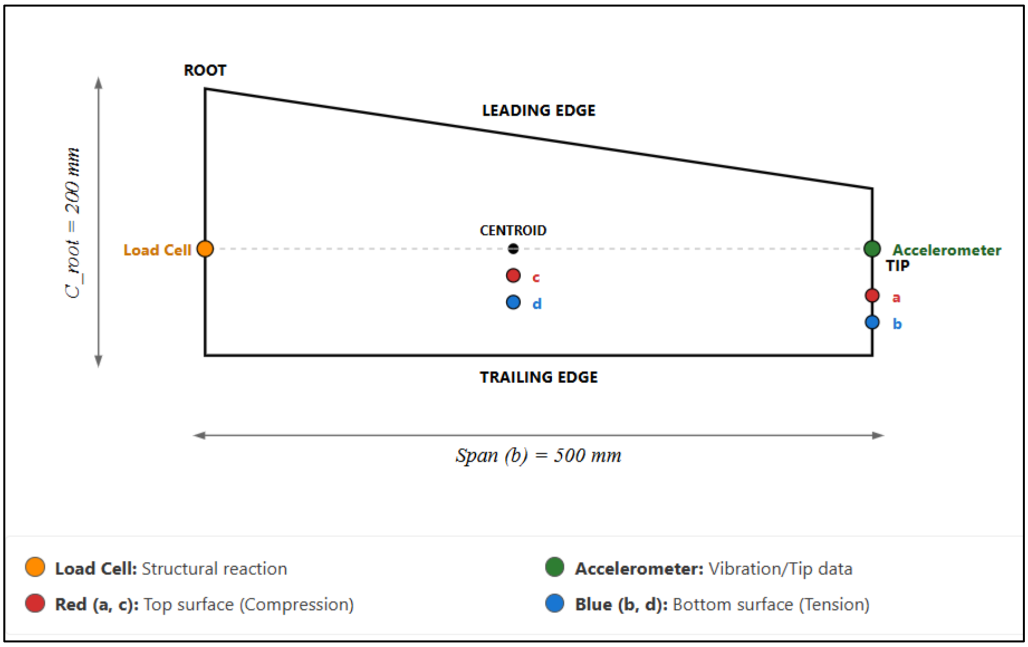


Figure 3.5: Strain Gauge Placement on Upper and Lower Wing Surfaces

The experimental set-up employed was the cantilever loading arrangement. The cantilever loading setup was such that the root of the wing was completely fixed and loads applied to the tip. The loading applied was increased from 0 kg to 5 kg in increments of 0.5 kg so that the amount of structural deformation at different increments was seen within the elastic range of the structure. The loading conditions employed during the experiment are displayed below in Table 3.4 and the full cantilever loading setup is shown in Figure 3.6.

Table 3.4: Experimental loading conditions

Load Step	Applied Load (kg)
1	0
2	0.5
3	1.0
4	1.5
5	2.0
6	2.5
7	3.0
8	3.5
9	4.0
10	4.5
11	5.0

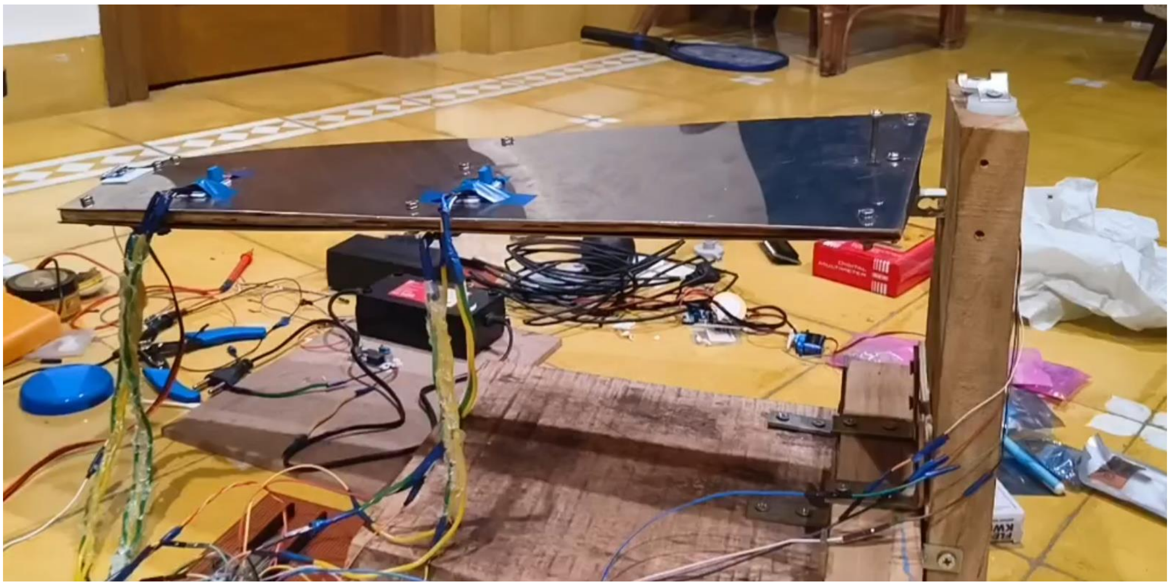


Figure 3.6: Experimental Setup Showing Cantilever Mounting and Tip Loading Arrangement

3.1 CFD analysis of the wing

An analysis of CFD was conducted using ANSYS Fluent to give an indication of the expected aerodynamic loading on the tapered wing configuration. While the purpose of the work is concerned with structural loading due to externally imposed forces, aerodynamic load conditions typical of this wing geometry were sought, to establish the pressure loading on the wing surface.

The computational domain was generated around the wing geometry and discretized using finite volume meshing techniques. Boundary layer refinement was incorporated near the wing surface to improve prediction of pressure gradients and near-wall flow behaviour. The k–omega turbulence model was selected because of its suitability for low-speed external aerodynamic applications and near-wall flow prediction. The data is listed in the Table 3.5. And the Figure 3.7 represents the Boundary Conditions of the fluid domain.

Table 3.5: CFD Simulation Parameters

PARAMETER	VALUE
TURBULENCE MODEL	k–omega
INLET VELOCITY	34.03 m/s
OPERATING PRESSURE	101325 Pa
SOLVER TYPE	Coupled
GRAVITY	Enabled

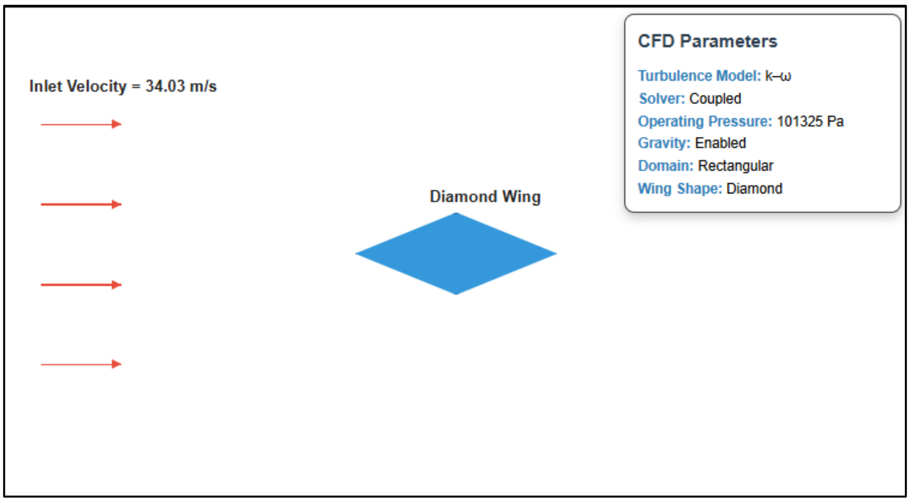


Figure 3.7: CFD Computational Domain and Boundary Conditions

Analysis CFD was conducted using ANSYS Fluent to give an indication of the expected aerodynamic loading on the tapered wing configuration. While the purpose of the work is concerned with structural loading due to externally imposed forces, aerodynamic load conditions typical of this wing geometry were sought, to establish the pressure loading on the wing surface.

The computational domain was created around the wing geometry and meshed using a finite volume method of meshing. Refinement of the near wall boundary layer was included to better predict the pressure gradients near the wing surface and the nature of the near-wall flow. The k-omega turbulence model was chosen, due to its appropriateness for external aerodynamic flows in the low-speed regime, and near-wall flow resolution. The Figure 3.8 and Figure 3.9 shows the Velocity and Pressure contour around the wing.

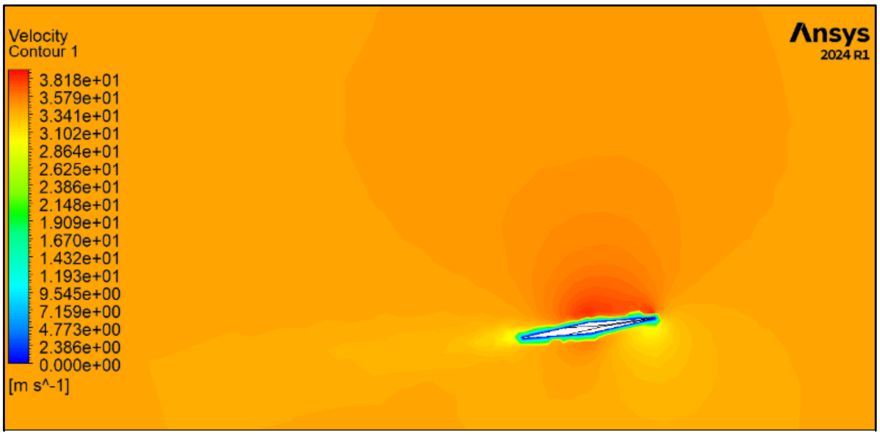


Figure 3.8: CFD Simulation Setup Showing Airflow Over Wing Geometry in Velocity contour

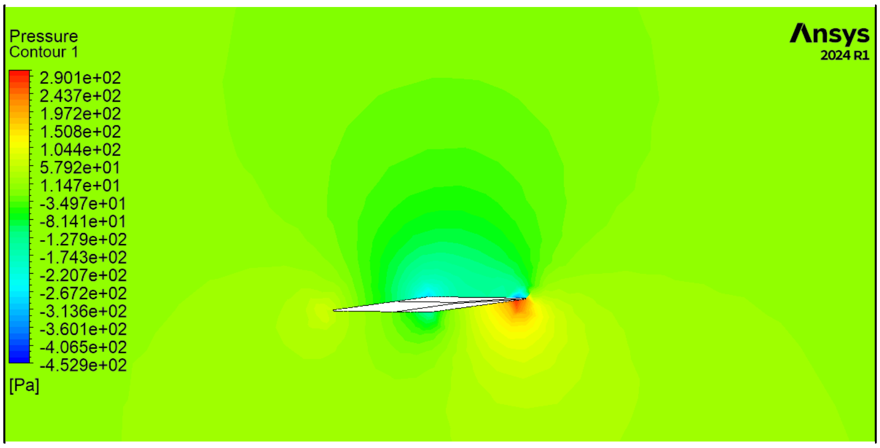


Figure 3.9: CFD Simulation Setup Showing Airflow Over Wing Geometry in Pressure contour

3.2 **FEM analysis of the wing**

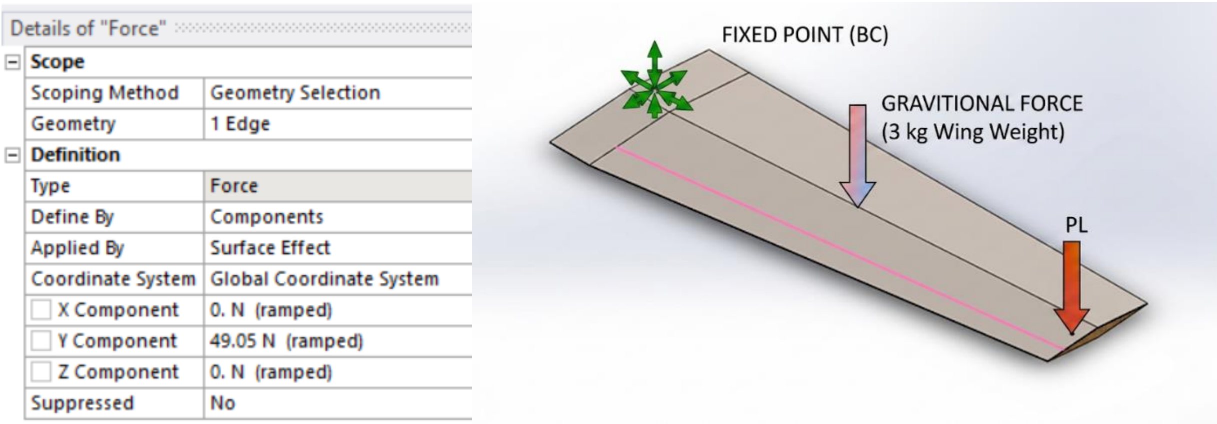
The structurally behaviour of the manufactured wing was calculated using a Finite Element Method (FEM) analysis conducted within ANSYS Mechanical with applied static cantilever loading. It was envisioned that through the application of numerical methods, stress concentrations, deformation patterns and spanwise response characteristics of the tapered sandwich wing could be determined and validated against experimentally obtained results from testing.

The finite element model was created in conjunction with actual dimension of the manufactured structure. The tapered wing section was modelled as a half-span of 500mm, root chord of 200mm and tip chord of 125mm with a thickness of 30mm. The structural composition of the wing included 304 stainless steel on the top and bottom face sheets with plywood material forming the core section. The numerical representation modelled the manufactured wing as close as is reasonably possible to its physical composition.

Static structural analysis conditions were selected for the FEM analysis due to the primary concern of determining deformation and stress responses under gradual external loading. To accurately model the cantilever mount present during testing the free end of the wing was free from any external supports and fixed supports were applied to the root end of the modelled structure. The external load that was applied to the free end of the wing was equivalent to the load used during physical testing and was added in discrete load increments. The Table 3.6: FEM Simulation ParametersTable 3.6 shows the details of the FEA model. The Figure 3.10 shows the Boundary conditions for the FEM model.

Table 3.6: FEM Simulation Parameters

PARAMETER	VALUE
ANALYSIS TYPE	Static Structural
STRUCTURAL CONFIGURATION	Tapered Sandwich Wing
LOAD RANGE	0–5 kg
LOAD INCREMENT	0.5 kg
BOUNDARY CONDITION	Fixed Support
OUTPUT PARAMETERS	Stress and Deformation



Details of "Force"	
Scope	
Scoping Method	Geometry Selection
Geometry	1 Edge
Definition	
Type	Force
Define By	Components
Applied By	Surface Effect
Coordinate System	Global Coordinate System
X Component	0. N (ramped)
Y Component	49.05 N (ramped)
Z Component	0. N (ramped)
Suppressed	No

Figure 3.10: Finite Element Model with Applied Boundary Conditions

The geometry of the wing structure was meshed in order to numerically determine the structural governing equations over the wing domain using finite element method. Mesh refinement was employed near the root section because bending moment varies from maximum value at the root, and higher stress gradients were predicted to exist around that region. This fine meshing provided better resolution to define localised stress concentration and localized deformation of the wing near the root section.

The cantilever beam theory governed the structural behaviour of the wing structure. A cantilever beam under transverse loading has a varying bending moment along the span, which is zero at the free end of the cantilever beam. Bending moment is maximum at the root of the cantilever beam and is given by:

$$\boldsymbol{M = Fx} \qquad \textbf{Equation 4}$$

Where, M is the bending moment (N), F is the applied force (N), and x is the distance from the free end (mm).

The normal bending stress present within the wing section is found using the simple flexure equation:

$$\boldsymbol{\sigma = \frac{Mc}{I}} \qquad \textbf{Equation 5}$$

Where, σ = bending stress (N/mm2 or MPa), M = bending moment (N-mm), c = distance from neutral axis (mm) and I = second moment of area (mm4). This equation signifies that bending stress is directly proportional to bending moment, c and inversely proportional to the second moment of area I. Hence, highest stress occurs at the uppermost and lowermost surfaces because c is largest at those sections.

The deflection behaviour of the cantilever wing structure was correlated using the Euler-Bernoulli relation for beams as stated below:

$$\boldsymbol{\delta = \frac{FL^3}{3EI}} \qquad \textbf{Equation 6}$$

Where, δ is the tip deflection (mm), F is the applied force (N), L is the length (mm), E is the young's modulus of the material (N/mm2) and I is the second moment of area of the section (mm4). This equation indicates that deflection is directly proportional to cube of span length and inversely proportional to the product EI, which signifies the stiffness of the structure. Because of the varying geometrical section of the wing, the EI was varying along the span. The finite element modelling took the consideration of varying EI.

The relation between stress and strain in the elastic region is given by Hooke's law:

$$\boldsymbol{\sigma = E\epsilon} \qquad \textbf{Equation 7}$$

Where, σ is the stress (MPa), E is the young's modulus (MPa) and \epsilon is the strain. This relation was used to correlate the finite element model with the strain readings obtained from the strain gauges mounted on the wing structure.

The resultant deformation was observed to be increasing with increasing load. Maximum deformation is observed near the free end because for a cantilever beam under transverse loading the highest deflection takes place at the free end of the beam. Contour plots of deformation clearly shown the smooth variation of deflection along the span. The Figure 3.11 and Figure 3.12 shows the deformation and the von-misses stress distribution on the wing in FEM, analysis on 5kg Point load + wing weight.

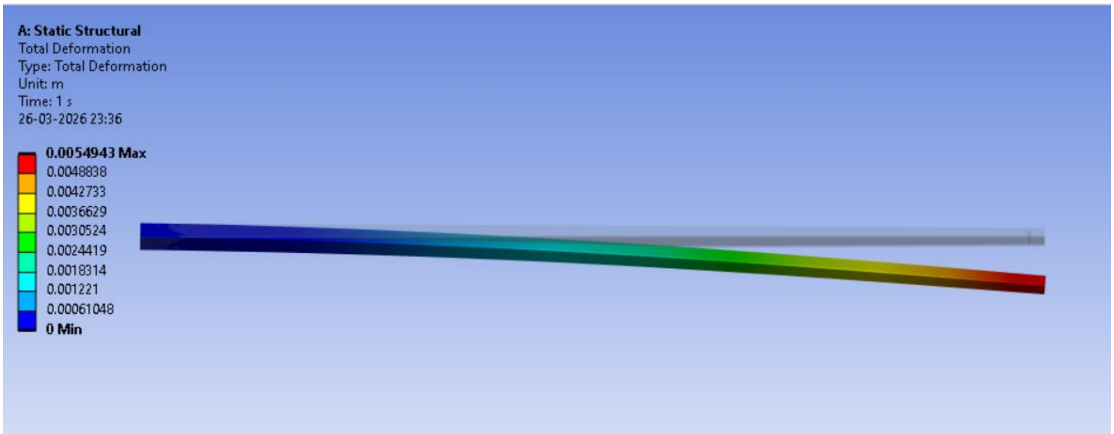


Figure 3.11: Total Deformation Contour Obtained from FEM Analysis

Contours representing the equivalent stress showed that higher stresses are observed near the root region and reduce toward the free end. This also shows that lower surfaces of the wing section undergo tension while the upper surfaces undergo compression when a transverse load is applied on the wing structure.

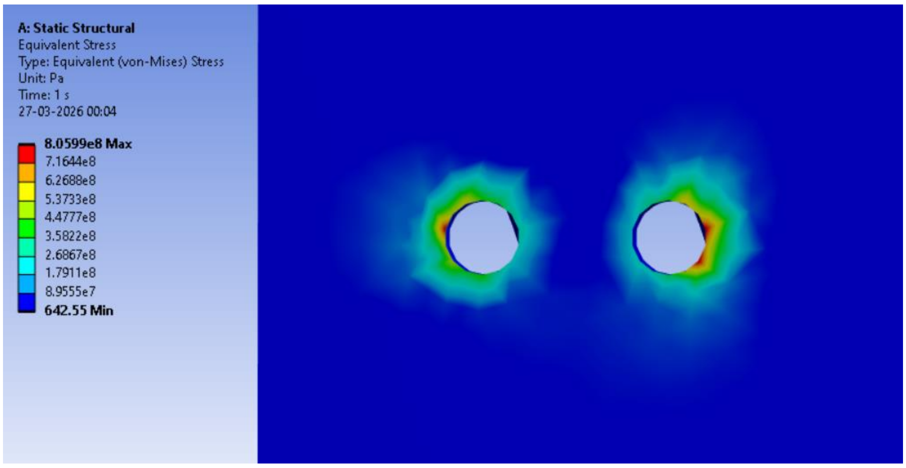


Figure 3.12: Equivalent Stress Distribution Contour Obtained from FEM Analysis

The analysis predicted approximately linear elastic response for load from 0 kg to 5 kg. Deformation was observed to be increasing linearly with increasing load, and there was no sudden stress concentration or failure within the entire load range studied. This confirmed the linear elastic behaviour of the wing structure during the experiment.

The resultant deformation and strain was correlated to the experimentally measured values. Numerical deformations predicted by FEM were somewhat lower than the experimentally measured deformations. The differences in the deformation are primarily due to the idealizations in the finite element model such as assumption of perfect rigid support conditions as compared to the relatively flexible support in the experiment. The minor differences might also be caused by manufacturing tolerances, slight non-uniformities in the adhesive layer, and non-uniformity of material properties in the fabricated sandwich structure.

However the trends observed in the experimental tests regarding deformations distribution, location of stress concentration and cantilever type behaviour have been well predicted by the finite element analysis. The finite element method can thus serve as a useful analytical tool in prediction of structural behaviour of such wing structures.

3.2 Summary of the simulation results for further study

This work aims to investigate the behaviour of a tapered composite wing under static cantilever loading and develop a correlation between experimental sensor data and analytical structural quantities. The response of a manufactured wing structure to increment loads and the conformity of these responses with analytically and numerically predicted values is investigated.

The design and construction of a scaled tapered wing with a root chord of 200mm, a tip chord of 125mm and 30mm thickness is among the primary aims of this work. The structure of the wing is that of a sandwich, with stainless steel 304 face sheets and a plywood core. The structure was chosen because it would offer adequate stiffness, was simple and could be manufactured at a low cost. The structure was to be able to tolerate a load of up to 5kg at the tip before suffering from plastic deformation.

To be able to analyse the behaviour of the structure, the wing needs to be fitted with sensors so as to be able to take readings. This involves the use of four strain gauges, two mounted on the upper surface of the wing and two on the lower surface at 50% and 100% of the wings span, so as to enable the calculation of tensile and compressive strains. The placement is selected to ensure that a measure of the difference in strain along the span can be analysed as well as compare to that which is expected. Also the applied load will be measured using a load cell, to ensure accurate application of a constant load and an accelerometer is added to obtain a value for tilt of the wing as another measure of deformation.

A significant portion of this work is the creation of a system that is capable of acquiring data from the sensors, and to communicate it with a processor system in a real-time manner, the microcontroller used to do this is an ESP32. Once data is received, this data will be processed so as to give useful structural parameters, such as deflection or strain, from the data from the sensors.

Another aspect of this work involves the analysis of how the response of the wing can be related to known structural values, this involves finding theoretical relations between bending stress, bending moment and tip deflection, based on beam theory. The values that were obtained by experiments using the sensors attached to the wing are put into the relevant formulas, so that comparisons can be made with analytical results.

As well as the experimental and analytical analysis, a numerical solution is also conducted, using ANSYS Mechanical, where it is possible to define all the properties of the wing structure and predict deflection and stress distribution. The analytical results are then compared to these simulated values. Aerodynamic force estimation is performed using ANSYS Fluent as another reference to analyze the effect of any potential loads during operation, though this is of secondary importance for this thesis compared to structural behavior.

CHAPTER 4

EXPERIMENTAL WORK & DATA ACQUISITION

4.1 Instrumentation and Data Acquisition System

To evaluate structural deformation and stress distribution and stresses generated within the wing section strain gauges are attached at two span wise locations, i.e., 50% span from the root and at 100% span from the root. At each location, strain gauges are mounted at both the top and bottom surfaces so as to read the compressive and tensile strains concurrently for the purpose of characterizing the bending behaviour. The strain gauge placement arrangement along the wing span is shown in Figure 4.1.

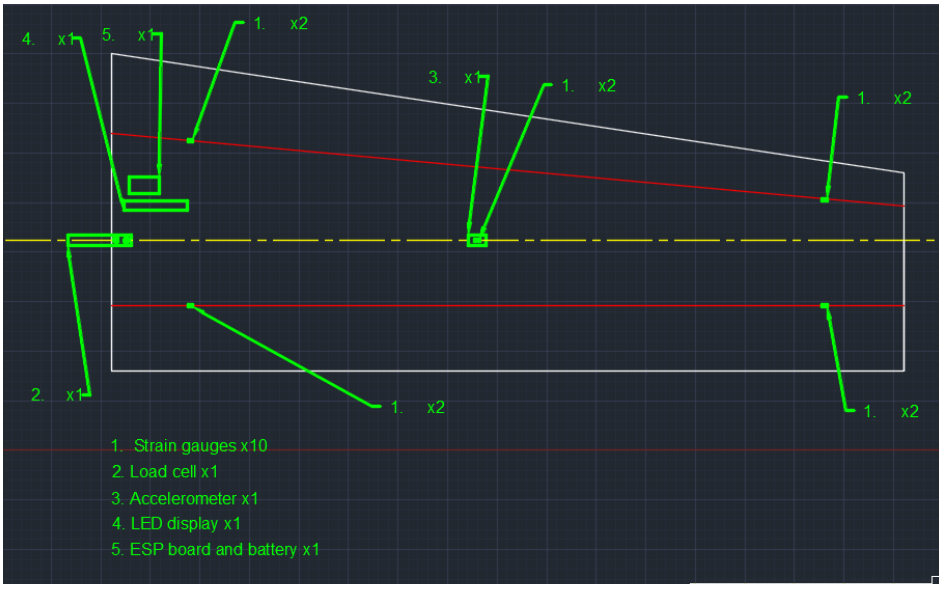


Figure 4.1: Strain Gauge Placement at 50% and 100% Span Locations on Upper and Lower Wing Surfaces

The two strain gauges at each location are wired to a data acquisition system, involving amplification and conditioning circuitry if required, to enable signal acquisition. The strain gauges are calibrated to measure strain in known stress level before carrying out experiments.

Load cell is used to measure the amount of load applied on the tip of the wing section. The load cell is calibrated to ensure accurate force measurement to convert the output of the cell to actual force, and it must be ensured that the full amount of the applied force pass through the sensing element of the load cell for measurement accuracy. The placement of the load cell at the tip loading location is shown in Figure 4.1.

To measure tilt angle of the wing, an accelerometer is used at tip section of the wing. Accelerometer reads the components of the gravity in x, y, z direction and from the measured values tilt angle with respect to gravity can be determined. The accelerometer mounting near the tip section is illustrated in Figure 4.1**.**

All signals from the strain gauges and the load cell (after some signal processing) along with tilt measurement from accelerometer is acquired by an ESP32 microcontroller. ADC channels of the microcontroller are used to record strain values while for the accelerometer, a serial interface is used. The overall data acquisition architecture used during experimentation is shown in Figure 4.2**.**

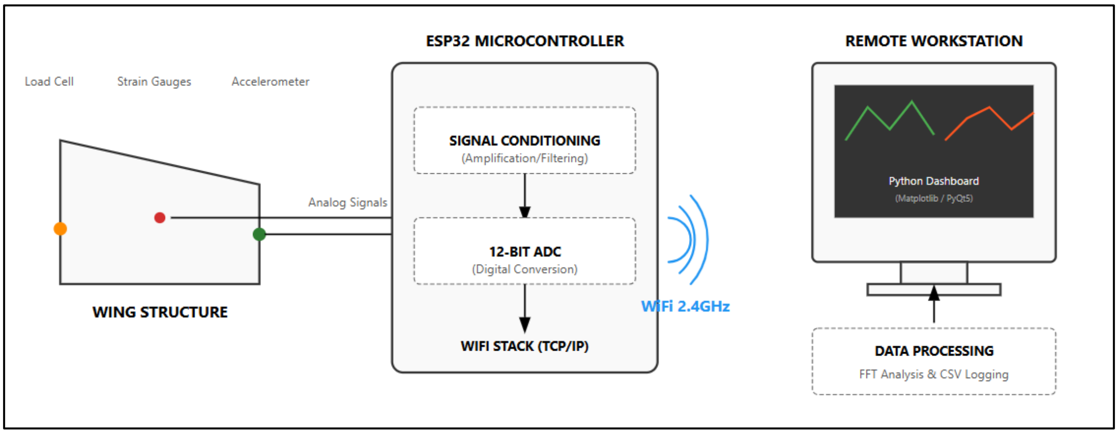


Figure 4.2: System Architecture for Experimental Data Acquisition and Processing

The measured output from the sensors are input to an ESP32 microcontroller, which was used as the data acquisition unit. The ESP32 reads the analog signals through its ADC channels. The sampling rate was set such that the response measured during incrementally applied load could be viewed as being almost quasi-static. The measured data was sent to the computer over the wireless link; each set of recorded data included strain at both top and bottom surfaces of the wing, load value and accelerometer output.

The data acquired by the microcontroller is transmitted to the processing unit via wireless means of HTTP. All readings are synchronized in order to simplify the process of obtaining various outputs. The components involved in the data acquisition and processing system are listed in Table 4.1**.**

Table 4.1: Components used for Data Acquisition

COMPONENT	FUNCTION
ESP32	Data acquisition
ADC	Signal conversion
WIFI	Data transmission
PYTHON	Data processing

The loading conditions used during experimentation are summarized in Table 4.2**.** Loads were progressively increased from 0 kg to 5 kg in increments of 0.5 kg in order to observe the variation of strain, stress, and deformation within the elastic region of the structure.

Table 4.2: Experimental loading conditions

LOAD STEP	LOAD (KG)	LOAD (N)
1	0	0
2	0.5	4.9
3	1.0	9.8
4	1.5	14.7
5	2.0	19.6
6	2.5	24.5
7	3.0	29.4
8	3.5	34.3
9	4.0	39.2
10	4.5	44.1
11	5.0	49.0

Finally, the experimental results were compared to those predicted analytically. The trends of the data are discussed and the difference between the experiment and the analysis is quantified. Issues such as the precision of boundary conditions are also discussed in detail and are used to evaluate how accurately the system represents the behavior of the structure. The Figure 4.3 shows the Cantilever loading setup used during the testing of the wing.

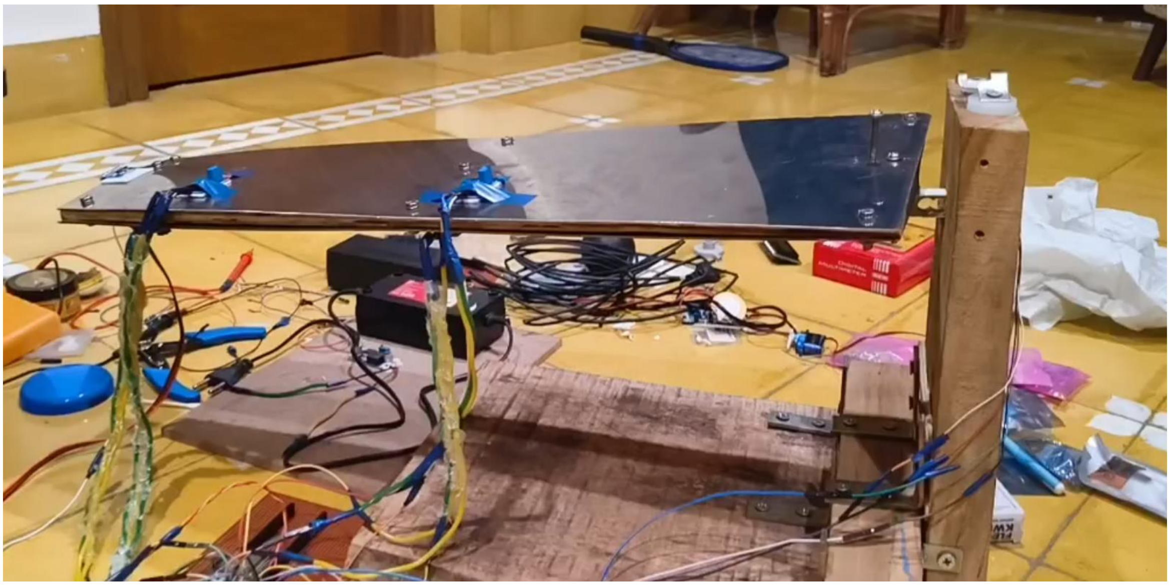


Figure 4.3: Experimental Cantilever Loading Setup Used During Testing

4.2 Analytical Framework and Structural Evaluation

To understand the response of the wing under static loading conditions the analytical relations are derived from classic beam theory. The wing structure is approximated as a cantilever beam loaded by a single concentrated load at its tip. This would cause bending along the span. The modeling as a cantilever is suitable for the current setup where only one load is applied and within elastic deformation range. The cantilever beam representation of the wing structure is shown in Figure 1.1**.**

Strain is used as the measurable parameter, whose value can be obtained from strain gauges mounted on the top and bottom surface of the wing. The stress induced due to strain is computed through Hooke's law.

$$\boldsymbol{\sigma = E\epsilon} \qquad \textbf{Equation 8}$$

Where σ is the stress, EEE is the elastic modulus of stainless steel 304, and ϵ\epsilonϵ is the strain reading obtained. Using this equation, stress developed at the position of the strain gauge can be computed.

The bending moment is determined for any spanwise section depending on the magnitude of load and the distance from the point of loading. Bending moment equation for a cantilever beam subjected to concentrated load at its free end is given as,

$$\boldsymbol{M = F \cdot x} \qquad \textbf{Equation 9}$$

Where FFF is the applied load, and xxx is the distance along the span to the location of the bending moment calculation. The bending moment varies linearly across the span and is maximum at the wing root. This accounts for the strain varying across the span. The variation of bending moment along the span is illustrated in Figure 3.11.

The stress induced due to bending is given by,

$$\boldsymbol{\sigma = \frac{Mc}{I}} \qquad \textbf{Equation 10}$$

Where ccc is the distance of extreme fiber from the neutral axis, and III is the second moment of area. III is calculated by using parallel axis theorem in order to consider both the behavior of face sheets and plywood core. An increase in the distance between face sheets makes larger III and hence increases bending stiffness. The sandwich section used for second moment of area calculation is shown in Figure 4.3**.**

The tip deflection is estimated using the cantilever beam relationship:

$$\Delta_{\boldsymbol{tip}} = \boldsymbol{\frac{FL^3}{3EI}} \qquad \textbf{Equation 11}$$

Where LLL is the span of the wing, and EIEIEI is the flexural rigidity. This equation allows estimation of the tip deflection. The deflected cantilever wing configuration is illustrated in Figure 3.12.

The aforementioned analytical relationships are used to analyze the measured raw data and present it in meaningful engineering quantities such as stress, moment, and deflection for comparison to the results from experimental and numerical analysis. The below Figure 4.4 shows the data integration of the IDE to the dashboard.

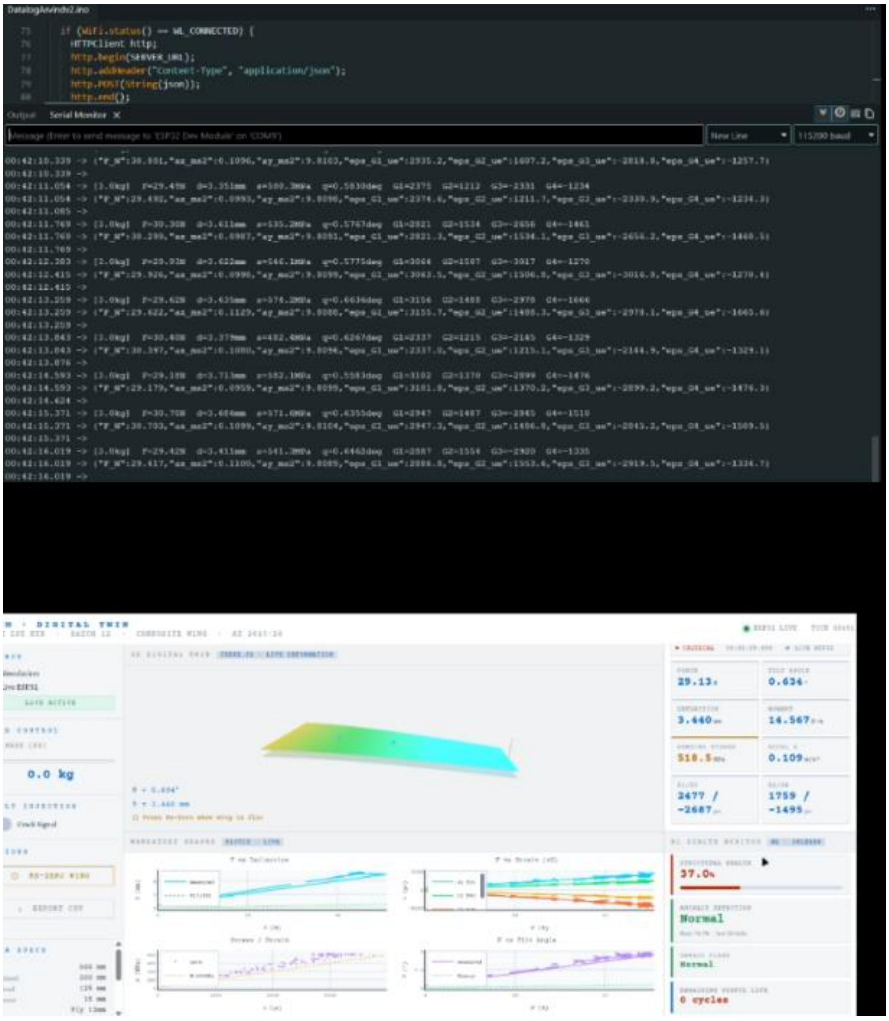


Figure 4.4: Data Acquisition System Showing Sensor Integration with ESP32 Microcontroller

4.3 Data Processing and Preparation for Results

The raw sensor output is processed to get the values of displacement, strain, stress and acceleration. Strain gauge values are converted to stress using Hooke’s law and deflection can be found out from the measured strain. The overall data processing sequence followed during the study is shown in Figure 4.4.

The data obtained is then organized to get the following results:

- Load vs Deflection
- Load vs Strain
- Stress vs Strain
- Load vs Tilt

These datasets are used in the next chapter to analyze structural behaviour, compare experimental and numerical results, and evaluate the accuracy of the system. The processed output parameters used for result generation are summarized in Table 4.3**.** And the final dashboard view is shown in the Figure 4.5.

Table 4.3: Processed Structural Response Parameters

PARAMETER	PURPOSE
LOAD VS DEFLECTION	Evaluation of wing stiffness
LOAD VS STRAIN	Observation of strain variation
STRESS VS STRAIN	Material behaviour analysis
LOAD VS TILT	Angular deformation evaluation

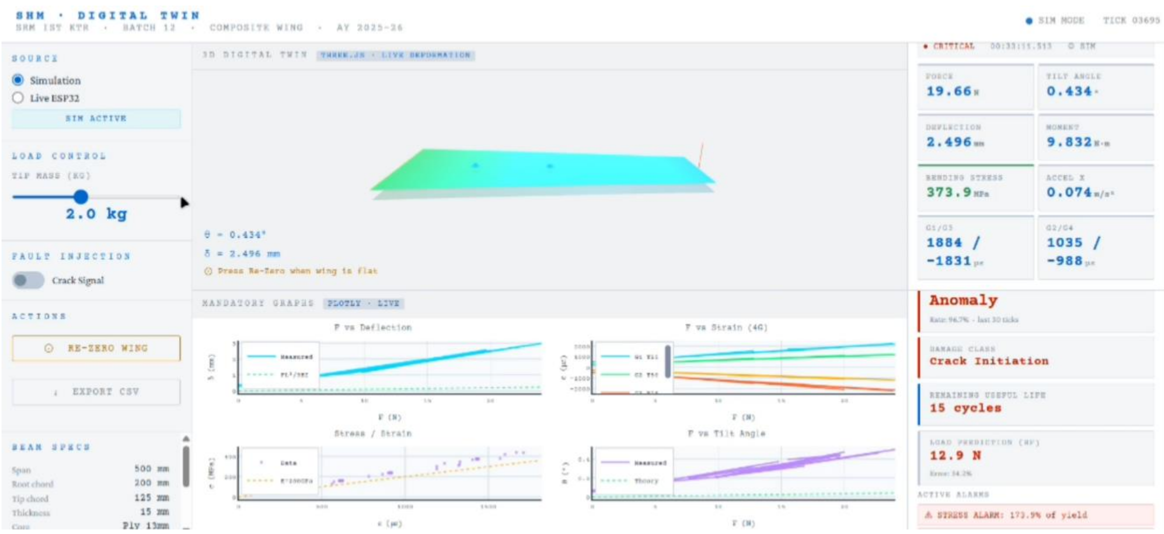


Figure 4.5: Processed Data Visualization Showing Structural Response Trends

CHAPTER 5

RESULT AND DISCUSSIONS

5.1 Overview of Experimental Observations

This chapter shows the static response of the wing when subjected to a load. The measurements used to analyze the static structural response were the values from strain gauges, the load cell and the accelerometer. The loads were incrementally applied from 0 kg to 5 kg and readings were taken at each step, when the measurements reached their stable value.

Using these readings, the relationships between the load and the deflection, strain, stress and tilt were plotted. These plots are analyzed to see if the wings response to a load follows expected trends according to cantilever beam theory. Also experimental readings are compared with analytical solutions and numerical simulations to evaluate consistency and determine differences.

5.2 Load vs Deflection Behaviour

This plot shows the relationship between the applied load and the deflection at the tip of the wing. The Figure 5.1 demonstrates the graph of Load vs Deflection curve comparing the experimental and analytical behaviour.

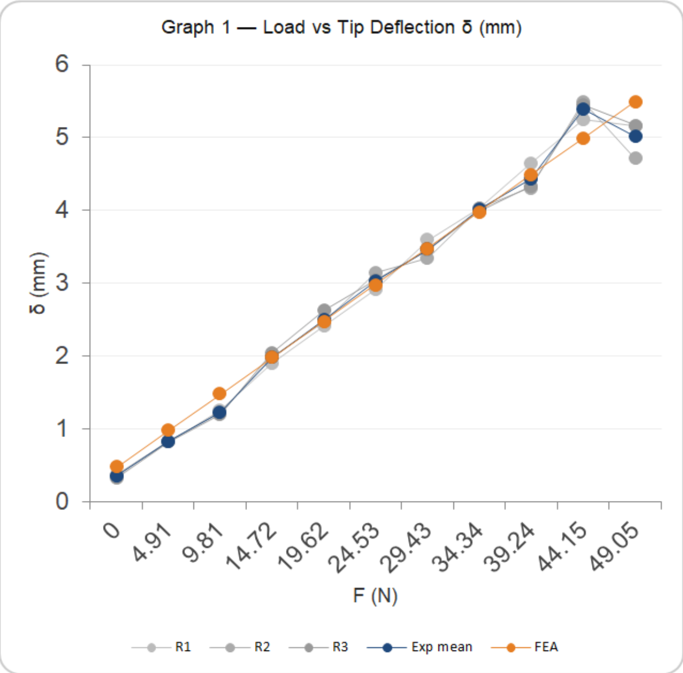


Figure 5.1: Load vs Deflection Curve Showing Experimental and Analytical Behaviour

From the plot, The results indicate that the load-deflection behavior is almost linear. This is consistent with theoretical predictions of the wing as expected, and is according to cantilever beam theory. According to this theory, within the elastic limit, the deflection of a beam is directly proportional to the load applied. The Table 5.1 shows the experimental results derived from the bending of the wing. In the results, the stress is a calculated value, else the deflection and the strain were directly acquired from the sensor readings.

Table 5.1: Experimental Results

LOAD (KG)	DEFLECTION (MM)	STRAIN (µE)	STRESS (MPA)
0.00	0.4739	94.78	90.07
0.50	0.9746	194.92	161.72
1.00	1.4761	295.22	233.24
1.50	1.9788	395.76	304.91
2.00	2.4805	496.1	376.43
2.50	2.9833	596.66	448.09
3.00	3.4851	697.02	519.62
3.50	3.9879	797.58	591.28
4.00	4.4897	897.94	662.80
4.50	4.9920	998.4	734.40
5.00	5.4943	1098.86	805.99

The linearity is observed until the loads were small and then there was a deviation in the linearity which could be caused by the compliance of the fixed boundary at the root. The boundary condition cannot be assumed as perfectly fixed. Thus some deformation occurs at the boundary rather than in the wing itself. As load is increased the effect of the compliance becomes less important. The analytical solution of the deflection at the tip is close to experimental values though there is some difference; This deviation is attributed to a less than perfect boundary condition and the accuracy of the experimental tests.

5.3 Load vs Stress Distribution

The above shows the change in stress along the span of the wing with varying load. The trend is typical for a cantilever beam, whereby the stress varies linearly with respect to load, meaning the structure is within the elastic limits for the entire range of load applied. These stresses were determined from the strain by use of hooke's law assuming the material behaves elastically. We can see that the stresses are maximal at the root section and decrease toward the wing tip. This is consistent with bending moments in a cantilever beam where the maximum bending moment occurs at the root and the bending moment linearly goes to zero at the tip, since bending stress is proportional to bending moment the stress along the span also decreases in the same way asbending moment distribution in cantilever beam. The experimental results are in good agreement with the theoretical predictions from the beam theory; they indicate predictable behaviour of structure.

Small discrepancy of stress could occur due to experimental errors or imperfect boundary conditions at the root section of the wing. The Figure 5.2 shows the graph of load vs stress.

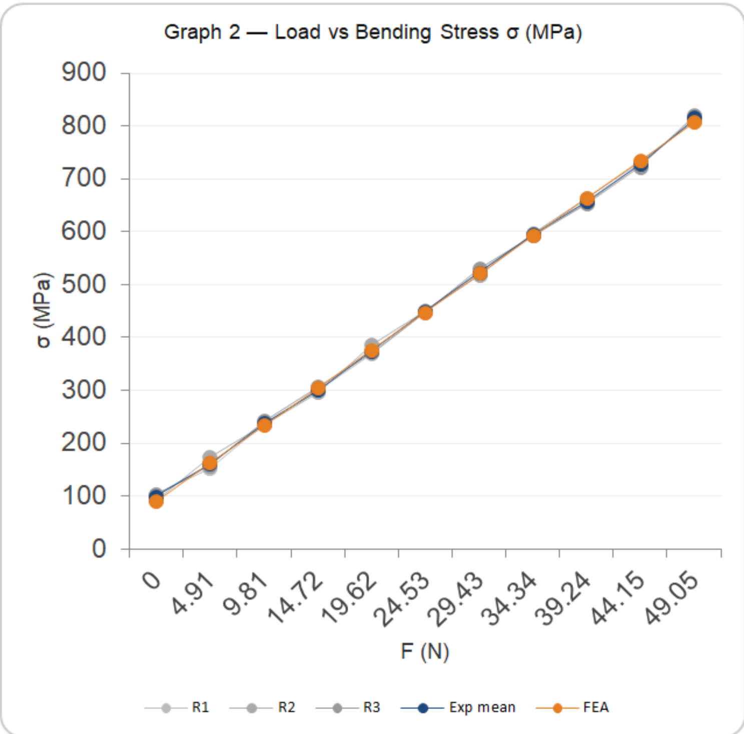


Figure 5.2: Load vs Stress Distribution

5.4 Stress–Strain Relationship

This plot shows the stress strain relationship of the material used to make the wing structure.

The results demonstrate that that the material behaves linearly and follows Hooke's Law for the entire duration of the test. Minor deviations are observed due to the expected linearity directly due to the difference in the readings for stress and strain at the same load level and also small differences in the reading of strain for both strain gauges on each surface due to small error in readings. However, it does show the linear elastic behaviour which is required to apply the theoretical equations. The Figure 5.3 represents the graph shows good compliance for the given test range.

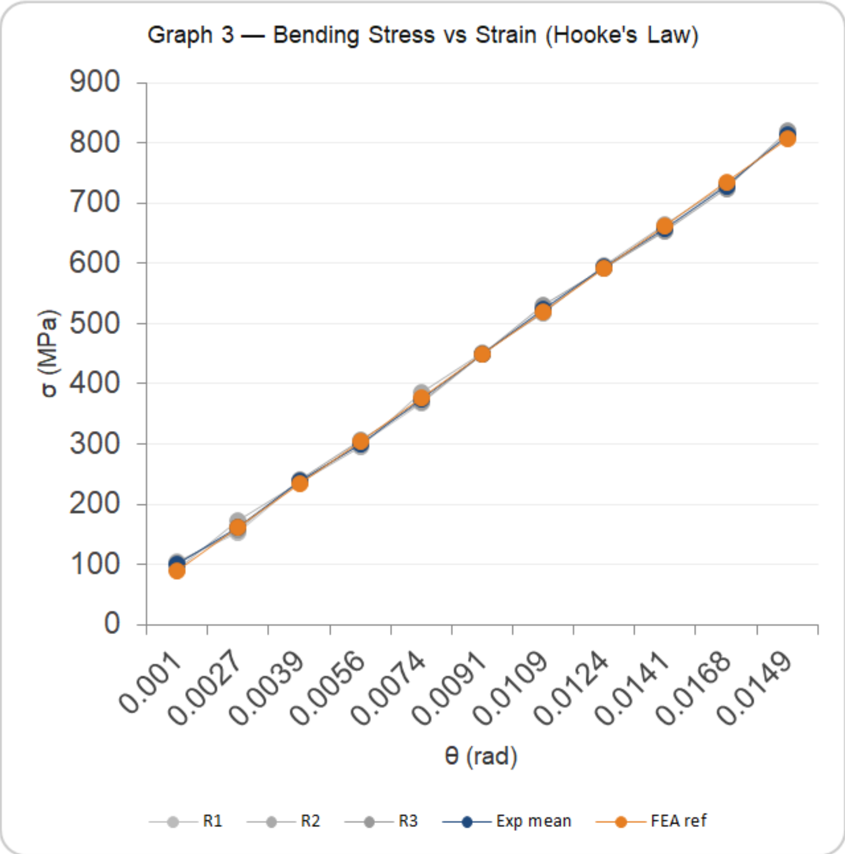


Figure 5.3: Stress–Strain Relationship Demonstrating Linear Elastic Behaviour

5.5 Load vs Tilt Behaviour

The results indicate that as the load is increased, the tilt angle is also increased. The graph is approximately linear but has slight inconsistencies, primarily due to measurement noise in the accelerometer when the readings are taken, especially noise when using the accelerometer and the measurement of the angle to the earth's horizon. However the trend does indicate the linear increase with load. The Figure 5.4 represents the plot that shows the measured tip tilt angle with each applied load.

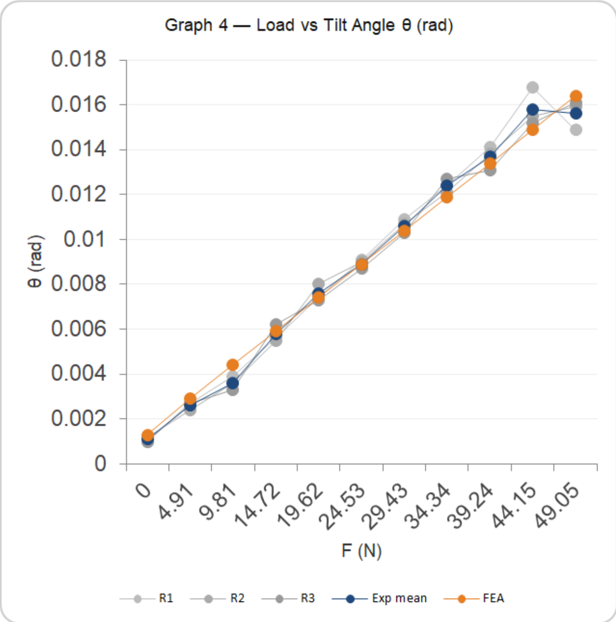


Figure 5.4: Load vs Tilt Angle Showing Angular Deformation of Wing

5.1 Comparison with Numerical Simulation (FEA)

In order to obtain additional information about the structure the finite element model of the structure was built in ANSYS. The following plots were created and compared with the experimental data. The Figure 5.6 and Figure 5.5 shows the comparison charts of FEM and Experimental results of Load vs tip deflection graph and Load vs Bending stress. The Table 5.2 and Table 5.3 shows the data of Deflection and stress for every load from FEM results and Experimental value respectively.

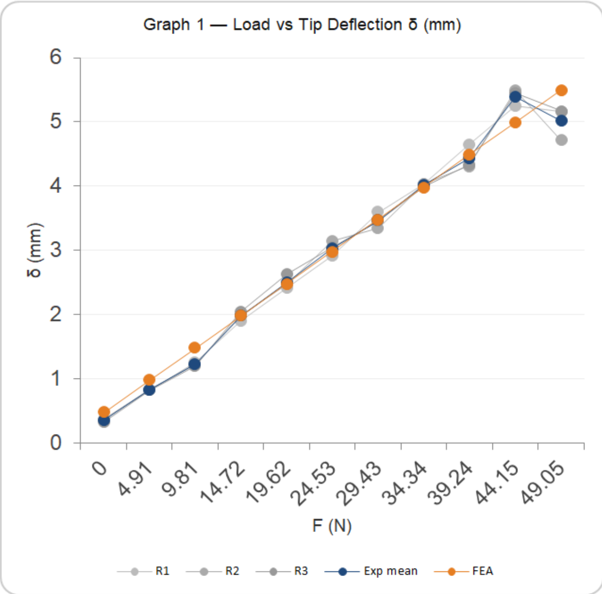


Figure 5.6: Comparison of Experimental and FEA Deflection Results

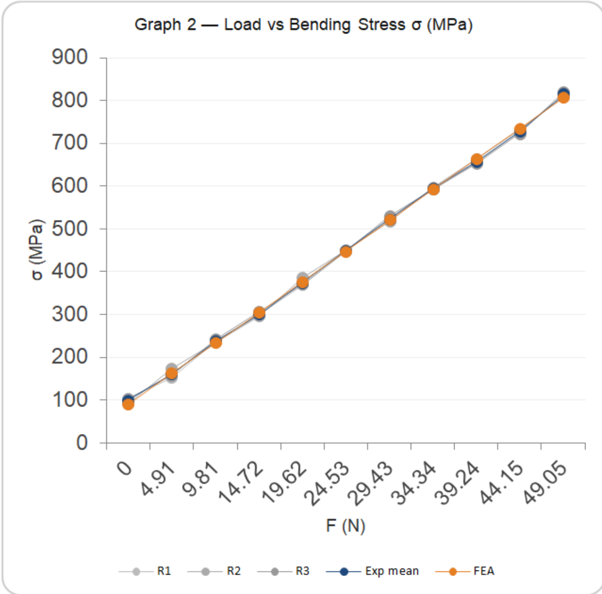


Figure 5.5: Comparison of Experimental and FEA Stress Distribution

Table 5.2: FEA Results done in ANSYS

LOAD (KG)	DEFLECTION (MM)	STRESS (MPA)
0.00	0.4739	90.07
0.50	0.9746	161.72
1.00	1.4761	233.24
1.50	1.9788	304.91
2.00	2.4805	376.43
2.50	2.9833	448.09
3.00	3.4851	519.62
3.50	3.9879	591.28
4.00	4.4897	662.80
4.50	4.9920	734.40
5.00	5.4943	805.99

Table 5.3: Comparison Table of Experimental and simulation data

LOAD (KG)	EXP DEFLECTION	FEA DEFLECTION	ERROR (%)
0.00	0.5116	0.4739	7.9
0.50	0.9492	0.9746	-2.6
1.00	1.6494	1.4761	11.7
1.50	2.0747	1.9788	4.8
2.00	2.5704	2.4805	3.6
2.50	3.0763	2.9833	3.1
3.00	3.5992	3.4851	3.3
3.50	4.1524	3.9879	4.1
4.00	4.6362	4.4897	3.3
4.50	4.9966	4.9920	0.1
5.00	5.5082	5.4943	0.3

The comparison shows that of deflection that the values calculated from FEA are lower than the measured values from experiments. This difference is attributed to the ideal boundary condition of a fixed constraint was imposed on the structure, which creates more stress in the material and thus a lower deflection than that predicted experimentally where a less than perfect constraint was applied to the structure. The stress in the material is more or less in the same agreement.

5.6 Aerodynamic Simulation Results (CFD)

The CFD analysis performed using ANSYS Fluent provides estimates of aerodynamic forces acting on the wing. The below Figure 5.7 shows the pressure contour over the wing done in ANSYS FLUENT software.

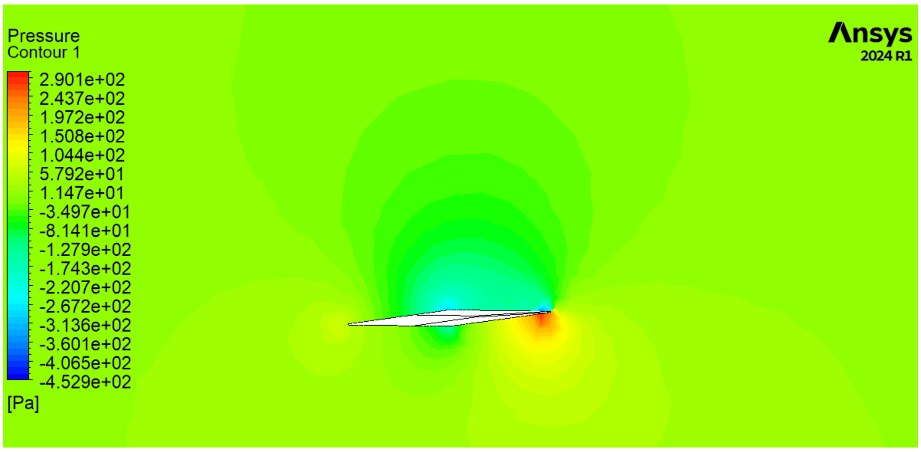


Figure 5.7: Pressure Contour Showing Aerodynamic Distribution over Wing Surface

Table 5.4: Aerodynamic Results

PARAMETER	VALUE
LIFT	-11.1054 N
DRAG	-408.742 N

An estimate of the loads that are imposed on the wing by aerodynamic forces was achieved by carrying out a computational fluid dynamics analysis on the profile using ANSYS. The analysis showed an average lift of -11.1054 N and drag value of -408.742 N. This analysis is not directly incorporated in this report but can be used to determine how much more force is imposed by aerodynamics that that achieved in the static tests reported here. These results can be used to estimate the stress imposed on the structure in operational conditions, as well as calculate lift and drag generated.

5.7 Data-Driven Model Performance

The regression model used for load prediction shows that strain measurements can be used to estimate applied load with reasonable accuracy. Table 5.5 shows the models used for prediction of the SHM.

Table 5.5: Machine Learning Models

MODEL	INPUT	OUTPUT
RANDOM FOREST	Strain	Load
ISOLATION FOREST	Sensor data	Anomaly

The regression model between strain and load is shown below. The model results indicate that The model is capable of predicting the actual load from the measurements of strain to a good degree of accuracy though at higher loads there appears to be some increase in the error between the predicted and actual load which is due to small fluctuations or noise in strain readings.

The anomaly detector plot show that there were no significant anomalies reported throughout the tests which confirms that the structural response during the experiment was consistent.

5.8 Discussion of Overall Structural Behaviour

From the graphs obtained in this investigation, The results indicate that the structure is linear in behavior as shown by the graphs of load vs deflection, load vs strain and load vs tilt angle. This confirms that Hooke’s law is applicable. The results further confirm that from comparison of analytical, numerical (FEA) and experimental results, that a linear relationship holds true between measured strain and applied load which can then be used to predict the actual load.

CHAPTER 6
CONCLUSION

An investigation into the static cantilever load response of a tapered composite wing using experimental tests, calculation and finite element analysis is presented. The test section is constructed from a sandwich composite made from stainless steel 304 face sheets with a plywood core. The test section has an overall span of 500mm and tapered geometry. The test section was instrumented with strain gauges, a load cell and an accelerometer, data was recorded as the load on the test section was incremented up to 5 kg.

Experimental results indicate that the test section behaved according to beam theory and within the elastic limit for the cantilever beam under cantilever load. This is demonstrated by the roughly linear load deflection response recorded by the test section, indicating that the test section was indeed acting within the elastic region. The response recorded from the strain gauges demonstrated that strain varies linearly with load as would be expected for elastic behavior. This, combined with the strain distribution readings along the span demonstrates that the bending moment is greatest at the root and decreases as the tip is approached. Hookes Law was determined to be valid for the test section through a linear relationship for stress-strain derived from measured results. Analytical results based on beam theory also coincided with the results obtained experimentally. The load-deflection curve of the experiment appears to give higher load deflections than the analytical calculations, this is due to inaccuracies in load deflection values owing to an imperfect clamped boundary condition at the root, and a degree of uncertainty from sensor readings.

The numerical results from ANSYS Mechanical mirrored the observed deformations and the distribution of stresses. A lower deflection value was obtained from ANSYS which is understandable given the perfectly fixed boundary condition implemented. The stresses obtained are consistent with what would be expected due to bending as the stress values are greatest at the root. A further test to look into the CFD results gives the airforces on the test section. These are not directly relevant for this analysis but provide an indication of loading conditions.

CHAPTER 7
FUTURE ENHANCEMENTS

The current work is limited to static loading only with simple assumptions on modeling. This study can be further extended to simulate the system under dynamic loading conditions i.e. Vibration analysis and fatigue analysis for a more realistic situation where the wings operate. The consideration of cyclic loading would be a crucial step towards establishing the long-term response and strength of the structures. Accuracy of the results could also be enhanced by accurately characterizing material properties, especially for the plywood core, and more precise imposition of the experimental and numerical boundary conditions.

Further improvements could also be made on the sensing and data analysis side. Increasing the number of strain gages over the span would give a better picture of the distribution of strains over the entire length of the wing while calibration of sensor and proper filtering could significantly reduce errors and noise. Aero-structural coupling could also be added to the numerical analysis such that the effect of load due to aerodynamics is analyzed over the wing structure and vise versa. Finally, advanced learning techniques could also be incorporated into data-driven module for accurate estimation of load and fault detection of potential issues in the wing structure for real-world monitoring application.

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